R V UNIVERSITY

School of Computer Science and Engineering

CASE STUDY REPORT

2020 Insider Social Engineering Attack on Twitter

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1. Case Study Overview

1.1 Objective

The main purpose of this report is to study a major social engineering incident and understand how it was handled using proper incident response and digital forensics practices. Instead of just explaining what happened during the incident, this report focuses more on how the organization recognized that something was wrong, how it reacted to the situation, how it controlled the damage, and how the investigation was carried out afterwards.

In this analysis, an Incident Response framework is used to examine the steps taken to detect the attack, contain its effects, and restore normal operations. The report also applies concepts from Digital Forensics to explain how digital evidence in such a case would be identified, preserved carefully, collected properly, and analyzed in a structured manner. This helps in understanding how investigators ensure that evidence remains reliable and legally valid during the investigation process.

By connecting theoretical knowledge from cybersecurity with a real-world incident, this study aims to show how these concepts are actually applied in practical situations. It also looks at the overall impact the incident had on the organization, including operational and reputational effects, and highlights areas where security controls and response strategies could be improved in the future.

1.2 Incident Summary

Type of incident:
This incident was an insider-enabled social engineering attack. In this case, the attackers did not attempt to hack into the company's public-facing systems through technical vulnerabilities. Instead, they focused on manipulating employees. By using social engineering techniques, they convinced certain employees to share their access credentials.

Once the attackers obtained these credentials, they were able to log into internal administrative tools that were meant only for authorized staff. Through these tools, they gained control over several high-profile accounts. This shows that the attack was not based on breaking security systems directly, but on exploiting human trust and internal access.

Target environment:
The attack mainly impacted the organization's internal corporate network, especially the account management and administrative support systems. These internal tools are normally used by employees to help users with issues such as resetting passwords, recovering accounts, and managing different account settings. They are designed to make support processes faster and more efficient.

However, these systems also have elevated privileges, meaning they provide higher levels of access compared to regular user accounts. Because of this, if someone gains unauthorized access to these tools, the consequences can be serious. An attacker can misuse these privileges to manipulate accounts, change sensitive information, or even take complete control of user profiles. This makes such internal systems highly sensitive and critical to protect.

Impact:
The consequences of the attack were both immediate and widespread:

- Verified and high-profile accounts were compromised
- Fraudulent cryptocurrency scam messages were posted publicly
- Approximately $100,000 worth of cryptocurrency was collected by the attackers
- Certain account features were temporarily restricted to prevent further misuse
- The organization faced reputational harm and increased regulatory attention

A key aspect of this incident is that no advanced malware or software vulnerability was exploited. The attackers relied on social engineering, specifically phone-based spear-phishing, to manipulate employees into revealing sensitive access information. This highlights an important cybersecurity reality: even strong technical systems can be weakened when human trust is exploited.

2. Incident Response Plan

2.1 Purpose

The purpose of an Incident Response Plan is to ensure that an organization reacts to a cybersecurity incident in a controlled and organized manner rather than responding in confusion or panic. When an incident occurs, there is often pressure to act quickly. A structured plan helps teams make decisions logically and in the correct sequence.

In this case, the Incident Response Plan serves as a guide for identifying the security breach, limiting its spread, removing the threat, and restoring normal operations. It also helps define who is responsible for each action, which reduces delays and miscommunication during critical moments.

An effective response plan aims to minimize operational disruption, financial loss, and reputational damage. It also ensures that evidence is preserved properly so that the incident can be investigated thoroughly. Overall, the plan provides a clear framework that supports both technical recovery and organizational stability during a cybersecurity crisis.

2.2 Incident Response Lifecycle

2.2.1 Preparation

Preparation is the foundation of any effective incident response process. It involves putting preventive measures, policies, and monitoring systems in place before an actual incident occurs. In this case, the organization had several important security controls already implemented within its enterprise environment.

These included multi-factor authentication (MFA), which requires users to verify their identity using more than just a password. The company also maintained logging and monitoring systems to track user activity, along with structured employee access management controls to regulate internal system privileges.

However, despite these measures, the principle of least privilege was not fully enforced. The principle of least privilege means that employees should only have access to the systems and data necessary to perform their specific job roles. In this situation, certain internal administrative tools were accessible to more employees than required, increasing the risk if credentials were compromised.

The primary weakness revealed during the incident was not a lack of technology, but insufficient protection against social engineering attacks targeting staff. While technical safeguards were present, there were gaps in preventing attackers from manipulating employees into revealing sensitive access information. This highlights that preparation must address both technical controls and human-centered risks.

2.2.2 Detection & Analysis

The incident was first noticed when several verified and high-profile accounts began posting similar cryptocurrency scam messages within a short period of time. The simultaneous nature of these posts raised immediate suspicion, as it suggested coordinated unauthorized access rather than individual account compromises.

Early indicators included unusual login activity associated with internal systems. Certain administrative actions were performed in a pattern that did not match normal employee behavior. There were also unexpected changes made to account settings, such as email modifications and password resets, which are typically controlled through internal support tools.

In addition to internal monitoring alerts, user complaints and public reports played a significant role in identifying the scale of the incident. Because the compromised accounts were highly visible, the suspicious posts were quickly noticed and reported.

A review of internal audit logs revealed that administrative support tools had been accessed without proper authorization. Log analysis showed that employee credentials were used to perform actions that aligned with the account takeovers. Further investigation determined that these credentials had been obtained through social engineering, specifically phone-based spear-phishing. This confirmed that the breach originated from compromised employee access rather than a direct technical vulnerability in the platform itself.

2.2.3 Containment

Once the unauthorized activity was confirmed, the primary goal shifted from investigation to immediate damage control. The focus during the containment phase was to stop the attackers from continuing to misuse internal tools and prevent further public impact.

The organization acted quickly to limit access and reduce the spread of the incident. The following immediate actions were taken:

- Locking all compromised accounts to prevent additional unauthorized posts
- Temporarily restricting access to internal administrative tools
- Forcing password resets for affected accounts and relevant employee credentials
- Disabling tweeting functionality for verified accounts for a short period

These steps were necessary to break the attackers' control over the system. By restricting internal tool access, the organization reduced the risk of further account takeovers. Locking accounts and resetting credentials ensured that previously obtained login information could no longer be used.

Although temporarily disabling verified accounts caused inconvenience to legitimate users, it was a strategic decision to prevent further misuse while the investigation continued. Overall, the containment measures were aimed at stabilizing the platform and restoring control over internal systems before moving to deeper corrective actions.

2.2.4 Eradication

After the immediate threat was contained, the next objective was to eliminate the root cause of the incident. While containment focused on stopping ongoing misuse, eradication aimed to ensure that the attackers could not regain access using the same method.

The organization conducted an internal review to identify exactly which employee accounts had been compromised. Once identified, those accounts were suspended or disabled to prevent any further unauthorized use.

The following corrective actions were implemented:

- Identification and suspension of compromised employee accounts
- Review and reduction of privileged access across internal systems
- Enforcement of stricter authentication requirements
- Enhanced monitoring of administrative and support tools

A detailed access review was performed to ensure that employees only retained permissions necessary for their roles. This reinforced the principle of least privilege, reducing the potential impact of future credential compromise.

Stronger authentication enforcement, including additional verification layers, was introduced to make unauthorized internal access more difficult. Monitoring of internal tools was also improved so that unusual administrative behavior could be detected more quickly in the future.

The eradication phase addressed not only the immediate compromised accounts but also the structural weaknesses that allowed the attack to succeed.

2.2.5 Recovery

Once the immediate threat was removed and internal systems were secured, the organization began restoring normal operations. The recovery phase focused on safely returning affected services to users while ensuring that the platform remained stable and secure.

The following recovery actions were carried out:

- Gradual restoration of access to compromised accounts
- Verification of system integrity to confirm no further unauthorized changes
- Transparent communication with affected users and the public
- Reinforcement of employee security awareness and training programs

Account restoration was done carefully rather than all at once, allowing security teams to monitor for any unusual activity during the process. System integrity checks were performed to confirm that internal tools, user data, and authentication systems were functioning correctly and had not been altered beyond the known impact.

Clear communication was also an important part of recovery. Informing users about what happened and what steps were taken helped maintain transparency and rebuild trust. In addition, internal security training was strengthened to reduce the likelihood of similar social engineering attacks in the future.

The recovery phase ensured that operations resumed in a controlled manner, while also reinforcing confidence in the platform's security measures.

2.2.6 Post-Incident Review

After the incident was fully contained and normal operations were restored, a detailed review was conducted to evaluate both the technical and procedural gaps that contributed to the breach. The purpose of this review was not only to understand what failed, but also to identify improvements that could reduce future risk.

The internal assessment revealed several key weaknesses:

- Excessive administrative privileges granted to certain employees
- Inadequate safeguards against insider manipulation through social engineering
- Limited real-time alerting for suspicious use of internal support tools

One of the most significant findings was that privileged access was broader than necessary. When too many users have elevated permissions, the impact of a single compromised account increases substantially. This highlighted the need to more strictly enforce the principle of least privilege.

The review also emphasized that technical security controls alone are not sufficient if employees can be manipulated into bypassing them. Stronger verification procedures for internal requests and additional employee awareness training were identified as necessary improvements.

Another important observation was the lack of immediate alerts specifically tailored to abnormal administrative behavior. Enhancing behavioral monitoring and implementing more detailed logging policies became a priority.

As a result of these findings, the organization developed recommendations focused on improving insider threat mitigation and adopting zero-trust security principles. Zero-trust is a security model that assumes no user or system should be automatically trusted, even if they are inside the network. Access must be continuously verified based on identity, context, and behavior.

The post-incident review transformed the event from a crisis into an opportunity for strengthening long-term security resilience.

2.3 Incident Response Diagrams & Screenshots

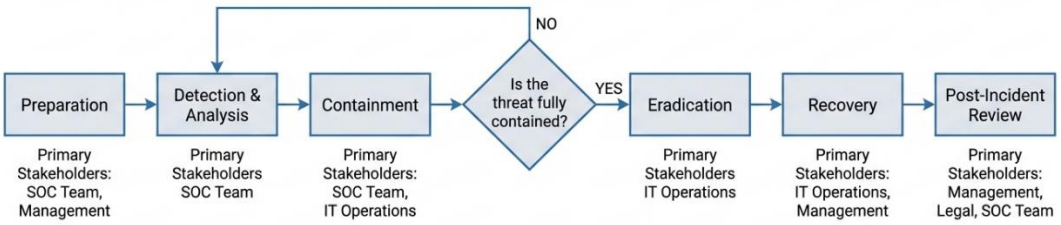

Figure 2.3.1: Incident Response Workflow

	Low Impact	Medium Impact	High Impact
Low Urgency	'Minor account misuse'	Limited internal access'	—
Medium Urgency	—	Verified account misuse	—
High Urgency	—	—	High-profile account compromise

Figure 2.3.2: Incident Classification Matrix

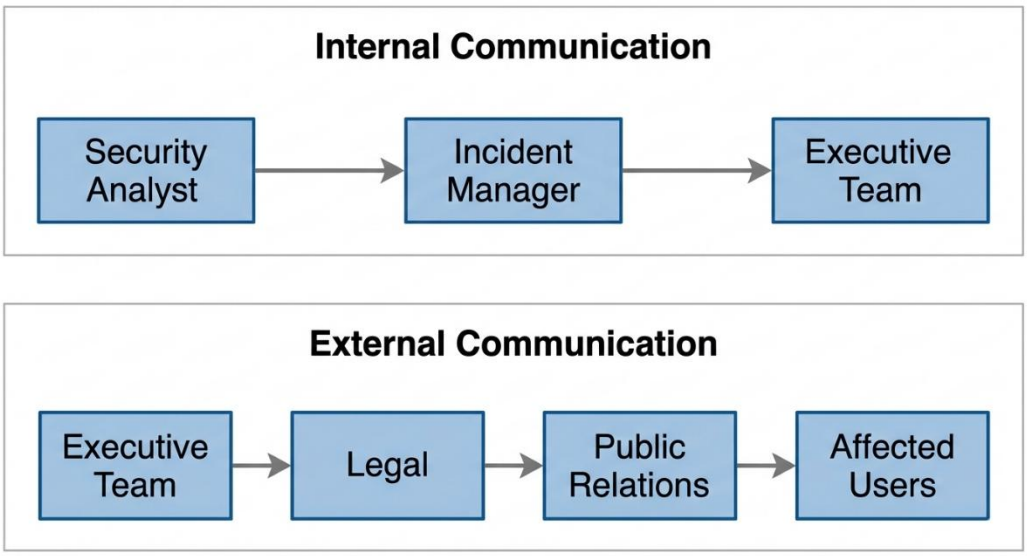

Figure 2.3.3: Communication Flow During Incident Response

2.4 Roles in Incident Response Report

Role	Responsibility
Incident Response Manager	Coordinates overall response strategy and decision-making
Security Analyst	Detects suspicious activity and conducts initial investigation
Forensic Specialist	Preserves and analyzes digital evidence
IT Operations Lead	Executes containment and recovery actions
Legal Advisor	Ensures compliance with data protection laws
Communications Officer	Manages public statements and media response
Documentation Officer	Records timelines, actions, and evidence

3. Digital Forensics Process

3.1 Purpose

The purpose of the Digital Forensics process is to investigate the incident in a structured and legally reliable manner. While incident response focuses on stopping and recovering from the attack, digital forensics focuses on understanding exactly how the breach occurred and collecting evidence that can support accountability.

Digital forensics involves identifying relevant digital evidence, preserving it so that it is not altered, examining it carefully, and documenting findings in a clear and defensible way. Maintaining integrity is essential. This means that the evidence must remain unchanged from the time it is collected until it is presented, and every step of handling must be recorded.

In this case, forensic analysis would aim to determine how employee credentials were obtained, how internal administrative tools were accessed, and what actions were performed using those tools. The findings could support internal disciplinary action, regulatory reporting, or even criminal investigation if required.

Overall, the digital forensics process ensures that technical investigation is conducted methodically rather than based on assumptions, and that the conclusions drawn are supported by verifiable evidence.

3.2 Digital Forensics Lifecycle

3.2.1 Identification

The first step in the digital forensics process is identifying potential sources of relevant evidence. This involves determining which systems, devices, and records may contain information related to the incident. In this case, since the breach involved misuse of internal administrative tools, the focus was primarily on internal enterprise systems rather than public-facing servers.

The following evidence sources were considered relevant:

- Administrative tool access logs
- Employee authentication logs
- Internal communication records
- Cryptocurrency wallet transaction traces

Administrative tool access logs are particularly important because they record actions performed within internal support systems. These logs can show which accounts were accessed, what changes were made, and at what time the activity occurred.

Employee authentication logs provide information about login attempts, session durations, IP addresses, and authentication methods used. These records help determine whether credentials were used from unusual locations or outside normal working hours.

Internal communication records, such as corporate messaging or email systems, may provide context regarding how employees were contacted or manipulated during the social engineering process.

Cryptocurrency wallet transaction traces can also serve as evidence. Even though cryptocurrency transactions are pseudonymous, blockchain records are publicly traceable. Investigators can analyze transaction flows to understand how the collected funds moved after the scam messages were posted.

The identification stage ensures that all potentially relevant digital artifacts are recognized early, reducing the risk of overlooking critical evidence.

3.2.2 Preservation

Once relevant evidence sources were identified, the next priority was to preserve them in their original state. Preservation ensures that digital evidence remains unchanged from the moment it is collected, which is essential for maintaining credibility and legal reliability.

The following steps were taken during this phase:

- Immediate isolation of affected systems
- Creation of forensic images
- Hash value generation (e.g., SHA-256) to verify integrity
- Careful documentation of evidence handling

Isolating affected systems prevents further alteration of logs or data. This may involve disconnecting devices from the network or restricting access to internal servers. The goal is to stop any ongoing activity that could modify evidence.

Instead of directly analyzing live systems, forensic images are created. A forensic image is an exact bit-by-bit copy of a storage device. This allows investigators to examine the copied data while keeping the original evidence untouched.

To confirm that the copied data remains unchanged, hash values such as SHA-256 are generated. A hash value is a unique digital fingerprint of a file or system image. If the hash value remains consistent, it proves that the data has not been altered.

All actions taken during preservation are recorded in detail. This documentation includes who handled the evidence, when it was accessed, and where it was stored. Proper documentation supports the chain of custody and ensures accountability throughout the investigation.

Preservation is a critical stage because any failure to maintain integrity can weaken the validity of the entire forensic process.

3.2.3 Collection

After ensuring that systems were preserved and isolated, the next step was the formal collection of digital evidence. Collection involves gathering relevant data in a controlled and documented manner so that it can later be examined without altering the original sources.

The following collection activities were performed:

- Secure disk imaging of affected employee systems
- Extraction of relevant log files from internal servers
- Preservation of volatile memory where applicable

Disk imaging was carried out on systems used by employees whose credentials were suspected to be compromised. This process creates a complete copy of the storage device, including deleted files and hidden system data. Collecting a full image allows investigators to perform deeper analysis later without repeatedly accessing the original system.

Log files from authentication systems and administrative tools were extracted in their original format. These logs contain timestamps, user actions, and system events that are essential for reconstructing a timeline of the incident.

In certain situations, volatile memory (RAM) may also be collected. Volatile memory stores temporary data such as active sessions, running processes, and encryption keys. Because this data disappears once a system is powered off, it must be captured carefully and quickly if deemed relevant.

Throughout the collection process, strict documentation was maintained to ensure that each piece of evidence was properly labeled, timestamped, and stored securely. This step builds the foundation for accurate examination and analysis in later stages.

3.2.4 Examination

After evidence was collected, the next stage involved examining the data in a structured and systematic way. The purpose of examination is to extract meaningful information from raw data without yet drawing final conclusions.

The following examination techniques were applied:

- Timeline reconstruction
- Log correlation analysis
- Access privilege review
- Authentication pattern evaluation

Timeline reconstruction involves arranging events in chronological order based on timestamps from logs and system records. By mapping out when credentials were used, when administrative tools were accessed, and when account changes occurred, investigators can clearly visualize how the incident unfolded over time.

Log correlation analysis refers to comparing multiple log sources to identify connections between events. For example, authentication logs may be matched with administrative tool activity logs to determine whether a specific login session directly led to account modifications.

An access privilege review is conducted to examine what level of permissions each compromised account had. This helps determine whether the actions performed were within the user's authorized capabilities or exceeded normal job responsibilities.

Authentication pattern evaluation focuses on identifying unusual login behavior. This may include logins from unexpected locations, unusual time patterns, or repeated access attempts. Such anomalies often indicate that legitimate credentials are being misused.

The examination phase transforms raw digital records into structured observations that can later support deeper analysis and final conclusions.

3.2.5 Analysis

During the analysis phase, investigators evaluated the findings from the examination stage to determine the root cause and overall scope of the incident. Unlike examination, which focuses on organizing and reviewing data, analysis involves interpreting that information to reach supported conclusions.

The forensic analysis confirmed the following:

- Employee credentials were compromised through social engineering
- Internal administrative support tools were accessed without proper authorization
- The account takeovers were coordinated rather than random

By reviewing authentication logs and internal communication patterns, investigators determined that legitimate employee credentials had been used to access sensitive systems. There was no evidence of malware exploiting system vulnerabilities. Instead, the data showed that attackers successfully manipulated employees into revealing access information.

Further analysis of administrative tool activity demonstrated that once access was obtained, the attackers used those tools to reset account details and post fraudulent messages. The timing and similarity of actions across multiple high-profile accounts indicated a planned and coordinated effort rather than isolated misuse.

This phase allowed investigators to confidently conclude that the breach resulted from credential compromise through social engineering, combined with elevated internal privileges that amplified the impact.

3.2.6 Reporting

After completing the analysis, the findings were compiled into a structured forensic report. The purpose of this report was to clearly document what occurred, how the investigation was conducted, and what conclusions were supported by the evidence.

The forensic report included:

- A detailed timeline of events
- A list of all evidence sources examined
- A description of the methods and tools used during the investigation
- The final conclusions based on verified findings

The timeline presented events in chronological order, starting from the initial credential compromise to the public posting of fraudulent messages and subsequent containment actions. This helped decision-makers understand how quickly the attack unfolded.

The evidence section described the logs, system images, authentication records, and transaction traces that were analyzed. Each source was documented carefully to maintain transparency.

The methods section explained how evidence was preserved, how hash values were used to verify integrity, and how logs were correlated to reconstruct the sequence of events. This ensures that the investigative process can be reviewed or repeated if necessary.

Finally, the conclusions summarized the confirmed root cause of the incident and clarified that the breach resulted from social engineering combined with excessive internal privileges. The reporting phase ensures that findings are not based on assumptions but on documented and verifiable evidence.

3.2.7 Presentation

The final stage of the digital forensics process involved formally presenting the findings to relevant stakeholders. This step ensures that the results of the investigation are clearly communicated and can be used for decision-making, compliance, or legal action.

The findings were presented to:

- Executive leadership
- Legal teams
- Law enforcement authorities

Executive leadership was briefed on the overall impact of the incident, the confirmed root cause, and the measures taken to prevent recurrence. This allowed management to make informed decisions regarding policy updates, public communication, and long-term security improvements.

Legal teams reviewed the evidence to ensure that documentation met regulatory requirements and that any disclosures were handled appropriately. Since high-profile accounts were involved, legal oversight was essential to address potential liabilities and compliance obligations.

Where applicable, findings were also shared with law enforcement authorities to support further investigation. Because the incident involved financial fraud and coordinated activity, properly documented digital evidence was necessary to assist potential criminal proceedings.

The presentation phase ensures that technical findings are translated into actionable information for organizational and legal response.

3.3 Digital Forensics Diagrams & Screenshots

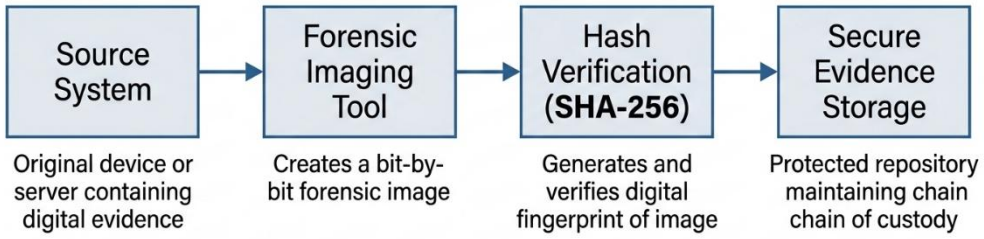

Figure 3.3.1: Digital Evidence Acquisition and Integrity Verification Process

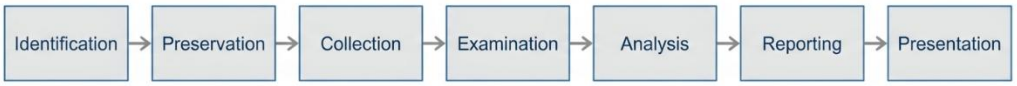

Figure 3.3.2: Digital Forensics Investigation Workflow

Evidence ID	Description	Collected By	Date/Time	Hash Value (SHA-256)	Storage Location
TW-IR-001 TW-IR-001	Forensic Image of Compromised Employee Workstation	Lead Forensic Investigator	15-07-2020 15-07-2020 09:45 AM	9f2a7c4d81e5b3a6...	Secure Evidence Vault – Case TW-2020
TW-IR-001	Image Accessed for Log Correlation Analysis	System Examiner	15-07-2020 01:30 PM	9f2a7c4d81e5b3a6...	Digital Forensics Lab – Workstation 2
TW-IR-001	Returned to Secure Storage	Evidence Custodian	15-07-2020 05:10 PM	9f2a7c4d81e5b3a6...	Secure Evidence Vault – Case TW-2020
TW-IR-002	Administrative Tool Access Logs (Internal Support System)	Forensic Analyst	16-07-2020 10:20 AM	4c8b2e9a73d5f1c0...	Secure Digital Repository – TW-2020 Logs
TW-IR-003	Authentication Server Login Records	Forensic Analyst	16-07-2020 11:05 AM	6a1d9f3c28b4e7d2...	Secure Digital Repository – TW-2020 Logs

Figure 3.3.3: Chain of Custody Record

3.4 Roles in Digital Forensics Report

Role	Responsibility
Lead Forensic Investigator	The Lead Forensic Investigator directs the overall investigation and ensures that all forensic procedures follow established standards. This role coordinates team activities and verifies that conclusions are supported by evidence.
Evidence Custodian	The Evidence Custodian is responsible for maintaining the integrity and secure storage of all collected digital evidence. This role ensures that the chain of custody is properly documented at every stage.
Malware Analyst	The Malware Analyst examines any suspicious files, scripts, or links related to the incident. This role determines whether malicious code was involved and evaluates its behaviour and impact.

Role	Responsibility
System Examiner	The System Examiner reviews system logs, user activity records, and device images to identify relevant artifacts. This role helps reconstruct technical events and supports timeline analysis.
Legal Compliance Officer	The Legal Compliance Officer ensures that the investigation follows applicable laws and regulatory requirements. This role reviews documentation to confirm that evidence handling meets admissibility standards.
Reporting Specialist	The Reporting Specialist prepares the formal forensic report based on verified findings. This role presents technical information in a clear and structured manner for stakeholders.
Quality Assurance Reviewer	The Quality Assurance Reviewer evaluates the investigation process to ensure accuracy and consistency. This role checks that methods used are reliable and that conclusions are logically supported.

4. Roles and Responsibilities

Incident Response and Digital Forensics teams serve different but closely connected purposes during a cybersecurity incident. Together, they ensure that the threat is controlled, operations are restored, evidence is preserved, and legal obligations are met.

The Incident Response (IR) team primarily focuses on managing the immediate impact of the attack. Their responsibilities include identifying the threat, containing its spread, removing the cause, and restoring affected systems. The main objective of IR is to reduce operational disruption and protect the organization from further damage.

In contrast, the Digital Forensics team concentrates on investigating the incident in depth. Their responsibility is to collect, preserve, and analyse digital evidence in a way that maintains integrity and supports accountability. Forensics ensures that conclusions are based on documented proof rather than assumptions.

Although their objectives differ, both teams must work collaboratively. For example, incident responders may need to isolate systems quickly, while forensic specialists must ensure that evidence is not accidentally altered during that process. Clear communication and documented procedures are therefore essential.

Maintaining accountability is critical throughout both processes. Each role must have clearly defined responsibilities, and every action taken during the incident should be recorded. This structured coordination ensures that technical response, investigation, and legal requirements are handled effectively and transparently.

5. Conclusion

The 2020 insider social engineering incident illustrates that even organizations with strong technical infrastructure remain vulnerable when human and procedural controls are not equally reinforced. The attack did not rely on complex malware or system vulnerabilities. Instead, it succeeded by exploiting trust and excessive internal privileges.

The incident response process was effective in limiting further damage and restoring operational stability within a short period. Immediate containment measures prevented additional account misuse, and recovery actions helped regain user trust. However, the investigation revealed weaknesses in access governance and monitoring practices that increased the overall impact of the breach.

Several important improvements were identified:

- Implementation of stricter least-privilege access controls
- Continuous monitoring of administrative tool usage

- Deployment of behavioural analytics to detect abnormal activity
- Regular social engineering awareness and simulation training
- Gradual adoption of zero-trust security principles

This case demonstrates that cybersecurity resilience depends on more than technical defenses alone. Strong policies, responsible access management, and employee awareness are equally important in preventing and responding to modern cyber threats. A balanced approach that integrates technology, governance, and human factors is essential for long-term security.

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